Answer Key - Chemistry - Graduate Level Assessment 11 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 102.5;*B*)100;*C*)109.5;*D*)106.78;*E*)101.22*Note:Thecorrectansweris*102.5ř*Cbecausethepresenceofsodium*

1. **Answer:** A
   Options: A) nitrogen; B) a halogenated hydrocarbon; C) chlorine; D) copper; E) arsenic
   Note: Nitrogen is a naturally occurring element that is essential for life and is not classified as a toxic pollutant in water, whereas many other substances can cause harmful effects at certain concentrations.

2. **Answer:** A
   Options: A) NaCl; B) LiCl; C) AlCl 3; D) KCl; E) FeCl 3
   Note: NaCl has the lowest melting point among the given compounds because it is an ionic compound with relatively weaker ionic bonds compared to the covalent bonds in other compounds, thus requiring less energy to break these bonds and transition to a liquid state.

3. **Answer:** A
   Options: A) Boiling point is 101.00,*Freezingpointis*1.55;*B*)*Boilingpointis*101.00,*Freezingpointis*−2.00;*C*)*Boilingpointis*99.57,*Freezingpointis*1.55;*D*)*Boilingpointis*100.0,*Freezingpointis*0.0;*E*)*Boil*

3. **Answer:** E
   Options: A) 10.0 g; B) 24.4 g; C) 15.2 g; D) 22.3 g; E) 19.6 g
   Note: The correct answer is 19.6 g because the weight of sulfuric acid is calculated using the formula: weight (g) = molarity (mol/L) × volume (L) × molar mass (g/mol), which in this case is 0.100 mol/L × 2.00 L × 98.1 g/mol = 19.62 g, rounded to 19.6 g.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because aluminum can exhibit multiple oxidation states, and in the case of AlBr 3, it has a +3 oxidation state, necessitating the use of a Roman numeral in its name to indicate this charge.

7. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because diluting a solution involves using the formula $C_1V_1 = C_2V_2$; substituting $C_1 = 3.50\,M$, $V_1 = 50\,ml$, and $C_2 = 2.00\,M$ yields $V_2 = 125\,ml$.

8. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because the expression simplifies to 1, as both the numerator and denominator are equal to 2 raised to the power of 0, which equals 1, thus demonstrating that the properties of exponents were misapplied.

9. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because the balanced decomposition reaction of sodium nitrate (NaNO 3) shows that 8.45 g of NaNO 3 yields 1.50 g of O_2 based on stoichiometric calculations and molar mass relationships.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because, according to the third law of thermodynamics, the zero-point entropy of a substance is not zero at absolute zero due to the presence of residual entropy associated with molecular motion and degeneracy in quantum states.

Long Form Response

11. **Answer:** To calculate the cell potential (E) for the electrochemical cell Ag/AgCl/Cl^-(0.0001)$and$$Ce^{4}$- Ce^3+(0.1)/Pt, we first identify the relevant half-reactions and their standard potentials. The standard reduction potential for Ag+/Ag is +0.799 V, and for Ce^4+/Ce^3+, it is +1.61 V. Using the Nernst equation, we account for the concentrations of $Cl^-$$and$$Ce^3$+, adjusting the potentials accordingly. After calculating the E for the cathode (Ce^4+ reduction) and the anode (Ag oxidation), we find the overall cell potential to be 0.9342 V, indicating the driving force for the electrochemical reaction.
    Note: correct because it accurately employs the Nernst equation to calculate the cell potential by incorporating the standard reduction potentials of the half-reactions and adjusting for the non-standard concentrations of the reactants, ultimately determining the overall cell potential based on the identified anode and cathode reactions.

12. **Answer:** The decomposition of solid mercury(II) oxide (HgO) into oxygen gas (O_2) and liquid mercury (Hg) can be represented by the balanced chemical equation: 2 HgO(s) $\rightarrow$ 2 Hg(l) + O_2(g). To calculate the moles of O_2 produced from 4.32 g of HgO, we first need the molar mass of HgO, which is approximately 216.59 g/mol. By dividing the mass of HgO by its molar mass, we find the number of moles: 4.32 g HgO $\times$ (1 mol HgO / 216.59 g HgO) = 0.0199 mol HgO. Given the stoichiometry of the reaction, 2 moles of HgO yield 1 mole of O_2, thus 0.0199 mol HgO will produce 0.0100 mol O_2. This calculation illustrates the relationship between the decomposition of HgO and the production of oxygen gas.
    Note: correct because it accurately applies stoichiometry and the molar mass of mercury(II) oxide to determine the moles of oxygen gas produced during its decomposition reaction, adhering to the balanced chemical equation.

End of Answer Key